Sankalp Jha

AI Engineer - Agentic Systems, LLM Orchestration & Backend
Delhi NCR, India

Portfolio | **LinkedIn** | **GitHub**
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Education

Year	Degree/Certificate	Institute	CPI/%
Oct 2022 – Aug 2026	B.Tech in CS & IT	Ajay Kumar Garg Engineering College, Delhi NCR	**7.25/10**
2021 – 2022	Senior Secondary (XII)	New Era School, Delhi NCR	**91.2%**

Experience

Software Engineer | *SYNRA Inc. , Tokyo, Japan* (***August 2026 – Present***)
RAG-Backed Analysis Platform & Cross-Platform Capture Client *Python, LangChain, FastAPI, Flutter, Swift, C++*

- **Retrieval Pipeline:** Built a RAG layer over vibration reports with FastAPI and LangChain.
- **Capture Client:** Shipped Flutter and iOS apps with locked exposure, focus and frame timing.

Google Summer of Code 2025 Contributor | *P4 Language Consortium, Stanford, CA* (***May 2025 – Sept 2025***)
SpliDT Project: Scaling Stateful Decision Tree Algorithms in P4 *P4, Python, P4Runtime, Docker, C++, Intel Tofino*

- **Switch-Native Architecture:** Compiled ML decision trees directly into P4 data planes with TCAM-efficient Subtree IDs to resolve hardware ASIC constraints.
- **Validation & Optimization:** Built a P4Runtime testing pipeline (Tofino/Mininet) improving reliability by **30%**; hit **10** Gbps line rate with **20%** lower inference latency.

Open Source Contributions

- **Project-HAMi/HAMi:** Fixed scheduler and device-plugin bugs in HAMi, a CNCF project for GPU/device resource sharing and isolation across Kubernetes clusters - **5** PRs merged.
- **WasmEdge/WasmEdge:** **#4470** fix(wasi): resolve **100%** CPU usage on macOS due to clock mismatch; **#4466** feat(docker): add standalone alpine-base image build pipeline.
- **facebook/bpfilter:** **#507** build: update clang-tidy brace config, fix opts.c/set.c, and NOLINT dump.c.

Key Projects

Cutable | *Go, Next.js, PostgreSQL, OpenRouter, E2B, WebSockets, Nomad* (*Project Link*)

Objective: Generate editable React applications from text and visual requirements.
Approach: Orchestrated OpenRouter tools and E2B sandboxes in Go; streamed code and live previews to Next.js over WebSockets.
Impact: Ships multimodal BYOK generation, persistent projects, and health-gated deployments.

DMIE | *Next.js, TypeScript, Python, FastAPI, Vector DB, LLM* (*Project Link*)

Objective: Determine whether a startup idea already exists by comparing it against more than **500,000** company records.
Approach: Built a scheduled scraper that keeps the startup dataset current; combined LLM query with vector search and backend filters.
Impact: Raised USD **10,000** and serves more than **100** users worldwide.

Moodish | *JavaScript, Node.js, OpenAI API, Swiggy MCP* (*Project Link*)

Objective: Cross-platform food recommendation app with personal and group-order flows.
Approach: Flutter Android and iOS clients across **6** tagged releases, with Google OAuth deep-linking and a live group-order lobby; Deterministic scoring over craving, budget and dietary constraints with an; inspectable recommendation trace.
Impact: Produces safer, inspectable food recommendations using Swiggy MCP

Myprod | *Go, Nomad, Traefik, WireGuard, Docker* (*Project Link*)

Objective: Turn manually provisioned VPS machines into a low-overhead personal compute pool for backend services.
Approach: Built a Go CLI and operator dashboard over Nomad scheduling, Traefik ingress, and WireGuard networking, with guarded node lifecycle controls and live resource metrics.
Impact: Provides one control surface for deploying services across heterogeneous free-tier and temporary compute.

Technical Skills

- **Languages:** Python, Go, TypeScript, JavaScript, C++, Dart, Swift
- **AI & Orchestration:** OpenRouter, Azure OpenAI, MCP, E2B, LLM tool-calling, vector search, Sklearn
- **Backend & Data:** FastAPI, NodeJS, gRPC, Protobufs, WebSockets, PostgreSQL, Redis, MongoDB
- **Tools / Platforms:** Docker, Nomad, CI/CD, Next.js, React, Linux, Kubernetes, eBPF, P4, Linux

Positions of Leadership

- **Technical Writer, Cloud Native Community Group, New Delhi** (***May 2024 – Dec 2024***)
  – Led **6+** writers and **4** designers on documentation and event content; social media revamp lifted engagement **60%**.
- **Co-Organizer & Anchor, Google Developer Groups & GDSC WOW 2024, Delhi NCR** (***Nov 2023 – Nov 2025***)
  – Co-led **10+** tech events; anchored GDSC WOW at IIIT Delhi (**800+** live audience, **31+** crew).

Publications

- R. Bharadwaj, **S. Jha**, V. Malyan, R. Gupta, "Trusture: A Blockchain-Based Decentralized Framework for Secure, Transparent, and Auditable NGO Transactions," ***2026 ICCSC***, Ghaziabad, India, Feb. **2026**. DOI: 10.1109/ICCSC67078.2026.11468597

Certifications & Honors

- **2x** AKTU State TT Gold (**2023**, **2024**); Robotics Olympiad Bronze, Thailand (**2019**); NASA, NPTEL & ISRO certifications